

Paper 1 - Set 06 - Solutions

Generated practice paper: answer key and worked solutions.

Answer key

Question	Answer	Topic
1	C	Quadratic functions
2	C	Integration
3	A	GCSE algebra
4	B	Indices and surds
5	D	Differentiation
6	C	Straight lines
7	B	Circles and circle theorems
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19	C	Triangles, radians and sectors
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Worked solutions

Question 1 QUA-05 · Quadratic functions · spec MM1.3

For what value of c does the line $y = 2x + c$ touch the parabola $y = x^2 - 4x + 7$?

Answer: C

Equate: $x^2 - 6x + (7 - c) = 0$. Tangency needs discriminant zero: $36 - 4(7 - c) = 0$, so $8 + 4c = 0$ and $c = -2$.

Full method - how and why

'Touches' means exactly one intersection, which after substitution means a repeated root: discriminant zero.

- 1 Set the two expressions equal: $2x + c = x^2 - 4x + 7$.

why: Intersections of $y = f(x)$ and $y = g(x)$ are the solutions of $f(x) = g(x)$.

- 2 Collect into a standard quadratic: $x^2 - 6x + (7 - c) = 0$.

why: Everything to one side, so the discriminant test can be applied.

- 3 Set the discriminant to zero: $(-6)^2 - 4(1)(7 - c) = 0$.

why: Touching means a repeated root, i.e. the line meets the curve once - exactly the $b^2 = 4ac$ case.

- 4 Solve: $36 - 28 + 4c = 0 \Rightarrow 4c = -8 \Rightarrow c = -2$.

why: Watch the double negative: $-4(7 - c) = -28 + 4c$.

Where it goes wrong

- Sign slip in $-4(7 - c)$ giving $c = 2$, which is offered.
- Using $b^2 - 4ac > 0$ (crossing) rather than $= 0$ (touching).

Takeaway Tangency between a line and a curve: substitute, then set the discriminant of the resulting quadratic to zero.

Question 2 INT-04 · Integration · spec MM7.6

A curve has $\frac{dy}{dx} = 6x^2 - 4x + 1$ and passes through $(1, 3)$. Find its equation.

Answer: C

$y = 2x^3 - 2x^2 + x + c$; at $(1, 3)$: $2 - 2 + 1 + c = 3 \Rightarrow c = 2$.

Full method - how and why

Reverse differentiation plus a boundary condition to pin down the constant.

- 1 Integrate each term: $\int(6x^2 - 4x + 1)dx = 2x^3 - 2x^2 + x + c$.

why: Add one to each index and divide: $\frac{6x^3}{3} = 2x^3$, $\frac{-4x^2}{2} = -2x^2$.

- 2 Do not forget the constant c - it is exactly what the given point determines.

why: Every antiderivative differs by a constant, so one point selects the right curve.

- 3 Substitute $(1, 3)$: $2 - 2 + 1 + c = 3$, so $c = 2$.

why: Evaluate the polynomial at $x = 1$ and set it equal to 3.

- 4 The curve is $y = 2x^3 - 2x^2 + x + 2$.

why: Check: differentiating gives back $6x^2 - 4x + 1$ ✓.

Where it goes wrong

- Omitting c altogether and choosing $y = 2x^3 - 2x^2 + x$.
- Arithmetic slip giving $c = 3$ - an offered option.

Takeaway $\frac{dy}{dx} = f(x)$ plus one point: integrate, then substitute to find c .

Question 3 ALG-04 · GCSE algebra · spec M4.8

Which of the following is an identity (true for all values of x)?

Answer: A

$$(x+3)^2 - (x-3)^2 = (x^2 + 6x + 9) - (x^2 - 6x + 9) = 12x \text{ for all } x.$$

Full method - how and why

An identity holds for **every** x . Expand each candidate and compare, or test values to eliminate.

- 1 Expand the first option: $(x+3)^2 - (x-3)^2 = (x^2 + 6x + 9) - (x^2 - 6x + 9)$.

why: Expand both squares fully, including the cross terms.

- 2 Simplify: the x^2 and 9 cancel, leaving $12x$ - true for all x . **This is the identity.**

why: Both sides agree identically, which is what 'identity' requires.

- 3 Check the others quickly: $(x+3)^2 = x^2 + 6x + 9 \neq x^2 + 9$; $(x+3)(x-3) = x^2 - 9$, not $+9$; $x^2 + 6x + 9 = 0$ holds only at $x = -3$; $(x-3)^2 = x^2 - 6x + 9$, not -9 .

why: Each wrong option drops or mis-signs a term - a quick test at $x = 1$ kills them all.

Where it goes wrong

- Confusing an equation (true sometimes) with an identity (true always).
- Forgetting the middle term when squaring a bracket.

Takeaway $a^2 - b^2 = (a-b)(a+b)$ applied to brackets gives quick identities; test $x = 1$ to eliminate impostors.

Question 4 IND-19 · Indices and surds · spec MM1.1

How many real solutions does the equation $2^{x^2} = 16 \times 2^{-3x}$ have?

Answer: B

$$2^{x^2} = 2^{4-3x}, \text{ so } x^2 + 3x - 4 = 0, \text{ i.e. } (x+4)(x-1) = 0: \text{ two real solutions.}$$

Full method - how and why

'How many real solutions' with exponentials - convert to a single base and the exponents give a quadratic whose discriminant answers the question.

- 1 Write $16 = 2^4$, so the right-hand side is $2^4 \cdot 2^{-3x} = 2^{4-3x}$.

why: Multiplying powers of the same base adds indices; this puts both sides in base 2.

- 2 Equate indices: $x^2 = 4 - 3x$, i.e. $x^2 + 3x - 4 = 0$.

why: Legitimate because 2^u is one-to-one.

- 3 Factorise: $(x+4)(x-1) = 0$ - two distinct real roots.

why: The question asks only for the *number* of solutions, so the discriminant ($9 + 16 > 0$) would do; factorising confirms both roots are real.

Where it goes wrong

- Rejecting $x = -4$ because it is negative - the **exponent** may be negative; it is 2^x that must stay positive.
- Reading 16×2^{-3x} as $(16 \times 2)^{-3x}$.

Takeaway Same base $\Rightarrow$ equate exponents; then it is an ordinary quadratic and every real root counts.

Question 5 DIF-01 · Differentiation · spec MM6.2

Find $\frac{dy}{dx}$ when $y = \frac{(3x-2)^2}{x^2}$.

Answer: D

$$y = 9 - 12x^{-1} + 4x^{-2}, \text{ so } y' = 12x^{-2} - 8x^{-3}.$$

Full method - how and why

The spec only allows differentiating powers of x , so the expression must first be written as a sum of such powers - expand and divide term by term.

- 1 Expand the numerator: $(3x-2)^2 = 9x^2 - 12x + 4$.
why: There is no quotient or chain rule available, so the fraction must be dismantled.
- 2 Divide each term by x^2 : $y = 9 - 12x^{-1} + 4x^{-2}$.
why: $\frac{9x^2}{x^2} = 9$, $\frac{-12x}{x^2} = -12x^{-1}$, $\frac{4}{x^2} = 4x^{-2}$.
- 3 Differentiate term by term with $\frac{d}{dx}x^n = nx^{n-1}$: $0 + 12x^{-2} - 8x^{-3}$.
why: $-12 \times (-1) = +12$ and $4 \times (-2) = -8$; the constant 9 differentiates to 0.
- 4 Write in fraction form: $\frac{12}{x^2} - \frac{8}{x^3}$.
why: Match the form used in the options.

Where it goes wrong

- Differentiating the bracket directly as $\frac{6(3x-2)}{x^2}$ - that would need the quotient rule (and is wrong anyway).
- Sign slips: differentiating x^{-1} gives $-x^{-2}$, so $-12x^{-1}$ gives $+12x^{-2}$.

Takeaway Rewrite as $\sum ax^n$ before differentiating; negative powers change sign when differentiated.

Question 6 LIN-01 · Straight lines · spec MM3.1

Find the equation of the line through $(2, -3)$ that is perpendicular to $3x - 2y + 5 = 0$.

Answer: C

The given line has gradient $\frac{3}{2}$, so the perpendicular has gradient $-\frac{2}{3}$:
 $y + 3 = -\frac{2}{3}(x - 2) \Rightarrow 3y + 9 = -2x + 4 \Rightarrow 2x + 3y + 5 = 0$.

Full method - how and why

Perpendicular line through a point: find the gradient of the given line, take the negative reciprocal, then use the point-gradient form.

- 1 Rearrange $3x - 2y + 5 = 0$ into $y = \frac{3}{2}x + \frac{5}{2}$, so its gradient is $\frac{3}{2}$.
why: Alternatively use the shortcut: $ax + by + c = 0$ has gradient $-\frac{a}{b} = -\frac{3}{-2} = \frac{3}{2}$.
- 2 Perpendicular gradient: $-\frac{1}{3/2} = -\frac{2}{3}$.
why: Perpendicular gradients multiply to -1 ; invert and change sign.
- 3 Use $y - y_1 = m(x - x_1)$ at $(2, -3)$: $y + 3 = -\frac{2}{3}(x - 2)$.
why: $y - (-3)$ is $y + 3$ - the commonest sign slip in this whole topic.
- 4 Clear fractions and tidy: $3y + 9 = -2x + 4 \Rightarrow 2x + 3y + 5 = 0$.
why: Multiply through by 3 first, then collect everything on one side.

Where it goes wrong

- Using $y - 3$ instead of $y + 3$, which gives $2x + 3y - 5 = 0$ - an offered option.
- Taking the perpendicular gradient as $-\frac{3}{2}$ or $\frac{2}{3}$.

Takeaway Negative **reciprocal**, and substitute the point with its own sign: $y - (-3) = y + 3$.

Question 7 CIR-01 · Circles and circle theorems · spec MM3.2

Find the radius of the circle $x^2 + y^2 - 6x + 4y - 12 = 0$.

Answer: B

$$(x - 3)^2 + (y + 2)^2 = 12 + 9 + 4 = 25, \text{ so } r = 5.$$

Full method - how and why

The expanded form of a circle. Complete the square in x and in y to recover the centre-radius form.

- 1 Group the terms: $(x^2 - 6x) + (y^2 + 4y) = 12$.

why: Move the constant to the right first so the completed squares have somewhere to go.

- 2 Complete each square: $x^2 - 6x = (x - 3)^2 - 9$ and $y^2 + 4y = (y + 2)^2 - 4$.

why: Halve each coefficient ($-6 \rightarrow -3$, $4 \rightarrow 2$) and subtract its square.

- 3 Substitute: $(x - 3)^2 + (y + 2)^2 - 13 = 12$, so $(x - 3)^2 + (y + 2)^2 = 25$.

why: The -9 and -4 move across to the right, **adding** to the constant.

- 4 Read off $r^2 = 25$, so $r = 5$.

why: The right-hand side is r^2 , not r .

Where it goes wrong

- Answering 25 - that is r^2 .
- Subtracting the 9 and 4 on the right instead of adding, giving $r^2 = 12$ or similar.

Takeaway $x^2 + y^2 + cx + dy + e = 0$ has centre $(-\frac{c}{2}, -\frac{d}{2})$ and $r^2 = \frac{c^2}{4} + \frac{d^2}{4} - e$.

Question 8 DIF-20 · Differentiation · spec MM6.3

Find the sum of the y -coordinates of the stationary points of the curve $y = 2x^3 - 9x^2 + 12x - 3$.

Answer: E

$y' = 6x^2 - 18x + 12 = 6(x - 1)(x - 2)$: stationary points at $x = 1$ ($y = 2 - 9 + 12 - 3 = 2$) and $x = 2$ ($y = 16 - 36 + 24 - 3 = 1$). Sum 3.

Full method - how and why

Find both stationary points and add their y -values - two substitutions after one factorisation.

- 1 Differentiate: $y' = 6x^2 - 18x + 12 = 6(x^2 - 3x + 2) = 6(x - 1)(x - 2)$.

why: Take out the 6, then factorise.

- 2 Stationary points at $x = 1$ and $x = 2$.

why: Roots of the derivative.

- 3 Compute the y -values: $y(1) = 2 - 9 + 12 - 3 = 2$; $y(2) = 16 - 36 + 24 - 3 = 1$.

why: Substitute into the original cubic, carefully: $2(8) = 16$ and $9(4) = 36$.

- 4 Add: $2 + 1 = 3$.

why: The question asks for the sum of the y -coordinates.

Where it goes wrong

- Adding the x -coordinates ($1 + 2 = 3$) - the same answer here by coincidence, but the wrong method.
- Arithmetic slips in $y(2)$; recompute $16 - 36 = -20$, $-20 + 24 = 4$, $4 - 3 = 1$.

Takeaway Two stationary points mean two substitutions into the original function - the derivative gives only the positions.

Question 9 SEQ-03 · Sequences and series · spec MM2.1

A sequence is defined by $x_{n+1} = 3x_n - 2$ with $x_1 = 2$. Find x_5 .

Answer: D

$$x_2 = 4, x_3 = 10, x_4 = 28, x_5 = 82.$$

Full method - how and why

A recurrence with a small index - just iterate. There is no shortcut worth finding for only four steps.

- 1 Apply the rule repeatedly from $x_1 = 2$: $x_2 = 3(2) - 2 = 4$.

why: $x_{n+1} = 3x_n - 2$ means 'triple the last term, subtract 2'.

- 2 $x_3 = 3(4) - 2 = 10$; $x_4 = 3(10) - 2 = 28$; $x_5 = 3(28) - 2 = 82$.

why: Write each step on its own line - the numbers grow quickly and a single slip propagates.

Where it goes wrong

- Stopping at $x_4 = 28$ (an offered option) by miscounting the steps: going from x_1 to x_5 takes **four** applications.
- Reading the rule as $3(x_n - 2)$, which gives $x_5 = 162$ - also offered.

Takeaway Count the applications carefully: reaching x_k from x_1 takes $k - 1$ steps.

Question 10 PRO-02 · Probability · spec M7.6

Two fair dice are thrown. Find the probability that the product of the two scores is odd.

Answer: C

The product is odd only if both scores are odd: $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$.

Full method - how and why

A product is odd only when **every** factor is odd - so this is an 'and' of two independent events.

- 1 Odd $\times$ odd = odd; any even factor makes the product even.

why: Recognising this converts the question into a much simpler one.

- 2 $P(\text{first die odd}) = \frac{3}{6} = \frac{1}{2}$; same for the second.

why: Three odd faces out of six: 1, 3, 5.

- 3 Independent dice, so multiply: $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$.

why: One die's outcome has no effect on the other.

Where it goes wrong

- Answering $\frac{1}{2}$ by thinking 'half of all products are odd' - in fact only 9 of the 36 outcomes are.
- Treating it as a **sum** (where odd + odd = even), a different and much more common question.

Takeaway Product odd $\Leftrightarrow$ all factors odd; product even $\Leftrightarrow$ at least one even.

Question 11 BIN-10 · Binomial expansion · spec MM2.4

Find the coefficient of x^5 in the expansion of $(1+x)^3(1+x)^4$.

Answer: B

$$(1+x)^3(1+x)^4 = (1+x)^7, \text{ and } \binom{7}{5} = 21.$$

Full method - how and why

Two binomials with the same bracket: combine the powers first, then use a single coefficient.

1 Combine: $(1+x)^3(1+x)^4 = (1+x)^7$.

why: Same base, so add the indices - this converts a product of expansions into one expansion.

2 The coefficient of x^5 is $\binom{7}{5}$.

why: Standard general term with $r = 5$.

3 Evaluate using symmetry: $\binom{7}{5} = \binom{7}{2} = \frac{7 \times 6}{2} = 21$.

why: Computing $\binom{7}{2}$ is quicker than $\binom{7}{5}$ and gives the same value.

Where it goes wrong

- Multiplying the separate expansions term by term and mis-collecting.
- Multiplying the coefficients $\binom{3}{5}$ and $\binom{4}{5}$ - both of which are zero, and neither is relevant.

Takeaway Combine equal brackets before expanding; then one binomial coefficient does the whole job.

Question 12 CIR-06 · Circles and circle theorems · spec MM3.3

A , B and C lie on a circle with centre O . Angle $AOB = 100^\circ$ and C lies on the major arc AB . Find angle ACB .

Answer: D

The angle at the centre is twice the angle at the circumference subtended by the same arc: $ACB = 50^\circ$.

Full method - how and why

The angle at the centre is twice the angle at the circumference on the same arc - provided you use the right arc.

1 Identify the arc: angle $AOB = 100^\circ$ at the centre stands on the **minor** arc AB .

why: The reflex angle at the centre (260°) would correspond to the other arc - which arc you use decides the answer.

2 C is on the major arc, so it sees the same minor arc AB .

why: The circumference angle must be on the opposite arc from the one it stands on.

3 Apply the theorem: angle $ACB = \frac{1}{2} \times 100^\circ = 50^\circ$.

why: Centre angle is double the circumference angle on the same arc.

Where it goes wrong

- Doubling instead of halving, giving 200° - impossible for an angle in a triangle.
- Using the reflex angle 260° and answering 130° (which would be right if C were on the minor arc).

Takeaway Angle at centre $= 2 \times$ angle at circumference **on the same arc**; check which side of the chord the point lies.

Question 13 FUN-01 · Functions and transformations · spec MM8.2

The graph of $y = x^2$ is translated by 3 units in the positive x -direction and then stretched parallel to the y -axis with scale factor 2. What is the equation of the resulting graph?

Answer: A

Translation: $y = (x - 3)^2$. Stretch by factor 2 in the y -direction: $y = 2(x - 3)^2$.

Full method - how and why

Two transformations applied in a stated order. Apply them one at a time to the equation, in the order given.

- 1 Translate 3 in the positive x -direction: replace x by $x - 3$, giving $y = (x - 3)^2$.
why: A shift **right** by a replaces x with $x - a$ - the sign is opposite to what it looks like, because the new graph reaches a given height 3 later.
- 2 Stretch parallel to the y -axis with factor 2: multiply the whole function by 2, giving $y = 2(x - 3)^2$.
why: A vertical stretch multiplies every y -value, so it acts on the outside of the function.

Where it goes wrong

- Writing $(x + 3)^2$ - shifting right uses $x - 3$.
- Putting the 2 inside as $(2x - 3)^2$, which is a horizontal stretch, not a vertical one.

Takeaway Changes **inside** the function affect x and act in the opposite sense; changes **outside** affect y and act as they read.

Question 14 LOG-03 · Exponentials and logarithms · spec MM5.3

Solve $2^{2x} - 3 \cdot 2^{x+2} + 32 = 0$. Find the sum of the solutions.

Answer: F

Let $y = 2^x$: $y^2 - 12y + 32 = (y - 4)(y - 8) = 0$, so $2^x = 4$ or 8 : $x = 2$ or 3 . Sum 5.

Full method - how and why

A hidden quadratic in 2^x . The middle term needs rewriting before the substitution is visible.

- 1 Rewrite the middle term: $3 \cdot 2^{x+2} = 3 \cdot 2^2 \cdot 2^x = 12 \cdot 2^x$.
why: $2^{x+2} = 2^x \times 4$ - separating the constant part is essential before substituting.
- 2 Substitute $y = 2^x$, noting $2^{2x} = y^2$: $y^2 - 12y + 32 = 0$.
why: $2^{2x} = (2^x)^2$; that is what makes it quadratic.
- 3 Factorise: $(y - 4)(y - 8) = 0$, so $y = 4$ or $y = 8$.
why: Both are positive, so both give solutions for x .
- 4 Convert back: $2^x = 4 \Rightarrow x = 2$; $2^x = 8 \Rightarrow x = 3$. Sum = 5.
why: The question asks for the sum of the x values, not the y values.

Where it goes wrong

- Summing the y values ($4 + 8 = 12$) - which is offered.
- Reading 2^{x+2} as $2^x + 2$.

Takeaway $a^{x+k} = a^k a^x$: split off the constant before substituting, then convert the roots back to x .

Question 15 GEO-15 · Angles, polygons, congruence and vectors · spec M5.2

A regular hexagon and a regular octagon share a common side. What is the size of the angle between their two other sides that meet at one end of the common side (the angle outside both polygons)?

Answer: C

Angles at the shared vertex: 120° (hexagon) $+ 135^\circ$ (octagon) $+ x = 360^\circ$, so $x = 105^\circ$.

Full method - how and why

Angles at a point sum to 360° . Find each polygon's interior angle and subtract.

- 1 Hexagon interior angle: exterior $= \frac{360}{6} = 60^\circ$, so interior $= 120^\circ$.
why: Regular polygon: interior $= 180 - \frac{360}{n}$.
- 2 Octagon interior angle: exterior $= \frac{360}{8} = 45^\circ$, so interior $= 135^\circ$.
why: Same formula with $n = 8$.
- 3 At the shared vertex three angles meet and fill a full turn: $120 + 135 + x = 360$.
why: The two interior angles plus the gap between the outer sides make a complete revolution.
- 4 Solve: $x = 360 - 255 = 105^\circ$.
why: That gap is the angle asked for.

Where it goes wrong

- Using 180° instead of 360° at the vertex.
- Using exterior angles in the sum, giving 255° or similar.

Takeaway Regular n -gon interior angle $= 180 - \frac{360}{n}$; angles round a point total 360° .

Question 16 INT-19 · Integration · spec MM7.1

The curve $y = x(x - 1)(x - 2)$ and the x -axis enclose two regions. What is the ratio of the area of the region for $0 \leq x \leq 1$ to the area for $1 \leq x \leq 2$?

Answer: E

The cubic has rotational symmetry about $(1, 0)$ (it is $u^3 - u$ with $u = x - 1$, an odd function), so the two regions are congruent: ratio 1 : 1.

Full method - how and why

A symmetry question. Shifting the cubic reveals it is odd about $(1, 0)$, so the two regions are congruent - no integration needed.

- 1 Substitute $u = x - 1$ to centre the cubic: $y = (u + 1)u(u - 1) = u(u^2 - 1) = u^3 - u$.
why: The roots 0, 1, 2 become $-1, 0, 1$ - symmetric about the origin, which is the hint to try this.
- 2 $u^3 - u$ is an **odd** function: replacing u by $-u$ negates it.
why: Odd functions have rotational symmetry of order 2 about the origin.
- 3 So the piece on $-1 \leq u \leq 0$ is the 180° rotation of the piece on $0 \leq u \leq 1$: the regions are congruent and have equal areas.
why: Rotation preserves area.
- 4 The ratio is 1 : 1.
why: You could integrate both (each has area $\frac{1}{4}$), but the symmetry argument is instant.

Where it goes wrong

- Assuming the regions differ because one is above the axis and one below - sign differs, area does not.
- Integrating carelessly and comparing signed values.

Takeaway Shift a cubic so its middle root is at the origin: if it becomes odd, the two enclosed regions are congruent.

Question 17 POL-09 · Polynomials, factor and remainder theorems · spec MM1.6

The expression $x^4 + 4$ can be factorised into two quadratics with integer coefficients. What is the sum of the coefficients of x in these two quadratics?

Answer: B

$x^4 + 4 = (x^2 + 2)^2 - 4x^2 = (x^2 + 2x + 2)(x^2 - 2x + 2)$. The x -coefficients are 2 and -2 , summing to 0.

Full method - how and why

$x^4 + 4$ has no real roots, so it cannot be split into linear factors - but it can be split into two quadratics via a difference of two squares.

1 Complete the square in x^2 : $x^4 + 4 = (x^2 + 2)^2 - 4x^2$.

why: $(x^2 + 2)^2 = x^4 + 4x^2 + 4$, which is $4x^2$ too big - so subtract it back. This manufactured square is the standard trick (the Sophie Germain identity).

2 Recognise a difference of two squares: $A^2 - B^2$ with $A = x^2 + 2$, $B = 2x$.

why: $4x^2 = (2x)^2$, so the expression is now in the form that factorises.

3 Factorise: $(x^2 + 2 - 2x)(x^2 + 2 + 2x) = (x^2 - 2x + 2)(x^2 + 2x + 2)$.

why: $A^2 - B^2 = (A - B)(A + B)$.

4 Add the x -coefficients: $-2 + 2 = 0$.

why: The question asks only for this sum, which the symmetry makes obvious.

Where it goes wrong

- Concluding it cannot be factorised because it has no real roots - a quartic with no real roots still factorises into two quadratics.
- Adding the constant terms ($2 + 2 = 4$) instead of the x -coefficients.

Takeaway $x^4 + 4a^4 = (x^2 - 2ax + 2a^2)(x^2 + 2ax + 2a^2)$: manufacture a square, then use the difference of two squares.

Question 18 GEO-11 · Angles, polygons, congruence and vectors · spec M5.19

$\vec{AB} = 3\mathbf{p} + 2\mathbf{q}$ and $\vec{BC} = k\mathbf{p} - 4\mathbf{q}$, where $\mathbf{p}$ and $\mathbf{q}$ are not parallel. Given that A , B and C are collinear, find k .

Answer: E

Collinear means $\vec{BC} = \lambda\vec{AB}$: $-4 = 2\lambda \Rightarrow \lambda = -2$, so $k = 3\lambda = -6$.

Full method - how and why

Collinear points mean one vector is a scalar multiple of the other. Compare coefficients of the two independent directions.

1 Collinear A, B, C means $\vec{BC} = \lambda\vec{AB}$ for some scalar λ .

why: They share the point B , so the two vectors must point along the same line.

2 Write it out: $k\mathbf{p} - 4\mathbf{q} = \lambda(3\mathbf{p} + 2\mathbf{q})$.

why: Because $\mathbf{p}$ and $\mathbf{q}$ are not parallel, coefficients can be compared independently.

3 Compare $\mathbf{q}$: $-4 = 2\lambda$, so $\lambda = -2$.

why: Use the equation that contains no unknown besides λ .

4 Compare $\mathbf{p}$: $k = 3\lambda = -6$.

why: Substitute the value of λ .

Where it goes wrong

- Taking $\lambda = 2$ and answering 6 - an offered option.
- Comparing $\mathbf{p}$ with $\mathbf{q}$ coefficients.

Takeaway Non-parallel vectors are independent: equate their coefficients separately to solve for scalars.

Question 19 TRI-07 · Triangles, radians and sectors · spec MM4.1

In triangle ABC , $AB = 6$, $BC = 4$ and angle $BAC = 30^\circ$. How many possible triangles are there?

Answer: C

Sine rule: $\frac{\sin C}{6} = \frac{\sin 30^\circ}{4}$, so $\sin C = \frac{3}{4}$. Since $BC = 4 < AB = 6$ and $4 > 6\sin 30^\circ = 3$, both the acute and obtuse values of C give valid triangles: the ambiguous case, two triangles.

Full method - how and why

Side-side-angle data: the ambiguous case. Compute $\sin C$, then decide whether both the acute and obtuse angles give a valid triangle.

- 1 Apply the sine rule: $\frac{\sin C}{AB} = \frac{\sin A}{BC}$, i.e. $\frac{\sin C}{6} = \frac{\sin 30^\circ}{4}$.
why: Pair each angle with the side opposite it: C is opposite $AB = 6$, and A is opposite $BC = 4$.
 $6 \times 0.5 = 3$
- 2 Solve: $\sin C = \frac{3}{4} = \frac{3}{4}$, which is less than 1, so solutions exist.
why: If $\sin C$ came out greater than 1, no triangle would exist at all.
- 3 Both $C = \arcsin \frac{3}{4}$ and $180^\circ - \arcsin \frac{3}{4}$ are possible in principle; check whether the obtuse one still leaves a positive third angle.
why: $\arcsin \frac{3}{4} \approx 48.6^\circ$, so the obtuse option is about 131.4° ; with $A = 30^\circ$ that totals under 180° ✓.
- 4 Equivalently: the side opposite the given angle (4) is shorter than the other given side (6) but longer than $6\sin 30^\circ = 3$ - the classic two-triangle condition. So there are 2 triangles.
why: The condition $b\sin A < a < b$ is the standard test for the ambiguous case.

Where it goes wrong

- Assuming one triangle because the sine rule 'gives one answer'.
- Assuming two triangles always - if $a \geq b$ the obtuse option would make the angles exceed 180° .

Takeaway SSA data: with $b\sin A < a < b$ there are two triangles; with $a \geq b$ exactly one; with $a < b\sin A$ none.

Question 20 IND-22 · Indices and surds · spec MM1.2

Which one of the following is the largest?

Answer: A

Each option is $\sqrt{a} + \sqrt{b}$ with $a + b = 9$. Squaring gives $9 + 2\sqrt{ab}$, which is largest when the product ab is largest, i.e. when a and b are as close as possible. Option E has $a = b = 4.5$ (product 20.25), beating $4 \times 5 = 20$ for option C. Numerically: $E = \sqrt{18} \approx 4.243$, $C = 2 + \sqrt{5} \approx 4.236$.

Full method - how and why

A comparison question. Every option has the form $\sqrt{a} + \sqrt{b}$ with $a + b = 9$ - spotting that shared structure turns guesswork into a one-line argument.

- 1 Notice that in every option the two numbers under the roots add to 9 (including $2\sqrt{4.5} = \sqrt{4.5} + \sqrt{4.5}$).
why: When all the candidates share a constraint, compare them through that constraint rather than by estimating each one.
- 2 Square a general option: $(\sqrt{a} + \sqrt{b})^2 = a + b + 2\sqrt{ab} = 9 + 2\sqrt{ab}$.
why: All the options are positive, so the largest option is the one with the largest square - and only the ab term varies.
- 3 With a fixed sum, ab is largest when a and b are closest together. The closest pair here is $a = b = 4.5$, giving $ab = 20.25$, beating $4 \times 5 = 20$.
why: This is the standard 'fixed perimeter, maximum area' fact; it decides the question without any decimals.
- 4 So $2\sqrt{4.5} = \sqrt{18} \approx 4.243$ is the largest (option C, $2 + \sqrt{5} \approx 4.236$, is a close second).
why: Worth a sanity check because the top two are so close.

Where it goes wrong

- Estimating roots to one decimal place - 4.24 versus 4.24 does not separate the top two options.
- Assuming $\sqrt{1} + \sqrt{8}$ is biggest because it contains the largest single number; the spread-out pair is actually the smallest.

Takeaway For a fixed sum $a + b$, the product ab - and hence $\sqrt{a} + \sqrt{b}$ - is maximised when a and b are as close as possible.